Highlights

SABRE: An Uncertainty-Aware Semantic–Behavioural Triage Framework for Adaptive Ransomware Detection

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- SABRE combines semantic and behavioural ransomware detectors within an uncertainty-aware modular framework.
- A triage policy uses predictive risk and epistemic uncertainty to support analyst escalation and automatic containment.
- Experiments on RDset and RanSMAP show that uncertainty-aware triage provides a conservative deployment mechanism under uneven evidence quality.

SABRE: An Uncertainty-Aware Semantic–Behavioural Triage Framework for Adaptive Ransomware Detection⋆

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ABSTRACT

Ransomware detection in operational environments is challenged by concept drift, behavioural polymorphism, evasion, and the cost of incorrect automated actions. In this paper, we present SABRE (Semantic-Behavioural Arbitration for Ransomware Evaluation), an uncertainty-aware modular framework for adaptive ransomware detection. The framework combines a semantic detector operating on Portable Executable (PE)-derived representations, a behavioural detector operating on time-window forensic telemetry, and a score-level decision layer that uses Monte Carlo Dropout-based uncertainty estimates to support risk-aware triage.

A central feature of SABRE is an interpretable triage policy defined by a risk threshold τ and an uncertainty threshold κ. High-risk, low-uncertainty samples may be automatically contained, whereas uncertain or borderline cases are escalated to human analysts. This design makes predictive uncertainty operational rather than purely descriptive. To support auditability, the framework also incorporates post-hoc explainability analyses including gradient saliency, permutation importance, and counterfactual perturbation analysis.

Experiments on RDset and RanSMAP show that the semantic and behavioural components exhibit substantially different discriminative characteristics, with near-saturated performance in the semantic setting and weaker separability in the behavioural setting. Across these settings, the main practical contribution of SABRE is a conservative uncertainty-aware decision process that links model outputs to analyst escalation and automated response in a transparent manner. Overall, the results support uncertainty-aware triage as a safer deployment mechanism for adaptive ransomware detection under evolving threat conditions.

1. Introduction

This section introduces the ransomware detection problem, motivates the need for uncertainty-aware decision support, and summarises the main contributions of SABRE.

Ransomware has become one of the most disruptive forms of cybercrime, with modern families exhibiting rapid mutation, behavioural polymorphism, and adaptive strategies designed to evade both traditional signature-based defences and contemporary machine learning detectors Fernando, Komninos and Chen (2020); Razaulla, Fachkha, Markarian, Gawanmeh, Mansoor, Fung and Assi (2023). These strains frequently employ obfuscation, polymorphism, and metamorphism to alter operational behaviour and code signatures, thereby undermining static detection techniques and increasing reliance on behavioural and AI-based methods Fernando et al. (2020); Urooj, Al-Rimy, Zainal, Saeed, Abdelmaboud and Nagmeldin (2024). Enterprise environments also operate under continuous concept drift, where benign and malicious behaviours evolve over time and trained models degrade unless they are retrained or recalibrated Ceschin, Botacin, Gomes, Pinage, Oliveira and Grégio (2022). Static classifiers that assume fixed data distributions therefore suffer rapid decay in effectiveness as attacker strategies change Fernando and Komninos (2024); Razaulla et al. (2023). These trends expose the limitations of monolithic ransomware detectors that lack explicit representations of uncertainty, adapt poorly to temporal distributional change, and provide limited support for calibrated human–AI interaction.

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∗Corresponding author
( ); ( )
( ); ( )
ORCID(s):
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Machine learning has been increasingly applied to ransomware detection, ranging from classical feature-based models to deep learning approaches that operate on system call traces, filesystem activity, API semantics, and other forensic telemetry Alraizza and Algarni (2023); Zahoora, Khan, Rajarajan, Khan, Asam and Jamal (2022). While such models can achieve strong in-distribution performance, they often behave unreliably on out-of-distribution (OOD) samples, novel ransomware variants, or previously unseen system behaviours Fernando et al. (2020). Most importantly, standard neural detectors provide point predictions without quantifying epistemic uncertainty, making it difficult to determine when the model is sufficiently reliable to support autonomous containment and when escalation to a human analyst is more appropriate Doan, Nguyen, Montague, Abraham, de Vel, Camtepe, Kanhere, Abbasnejad and Ranasinghe (2024). This absence of uncertainty-awareness leads to undesirable trade-offs: aggressive thresholds may cause costly false positives, whereas conservative thresholds increase analyst workload and delay incident response Ispahany, Islam, Islam and Khan (2024).

To address these challenges, we propose SABRE (Semantic–Behavioural Arbitration for Ransomware Evaluation), an uncertainty-aware modular framework for adaptive ransomware detection. Rather than relying on a single detector, SABRE uses two specialised components: a semantic detector trained on PE(Portable Executable)-derived representations and a behavioural detector trained on runtime telemetry. Each component produces both a predictive score and an epistemic uncertainty estimate via Monte Carlo Dropout. A score-level decision layer then combines these outputs and applies a triage policy based on a risk threshold τ and an uncertainty threshold κ. This policy allows automatic containment only when the predicted risk is sufficiently high and the estimated uncertainty is sufficiently low; otherwise, uncertain or borderline cases are escalated to human analysts. In this sense, the main contribution of SABRE is not merely predictive modelling but the operationalisation of uncertainty as part of a transparent response policy for ransomware defence.

Compared with prior ransomware detection studies, the novelty of SABRE lies in three aspects. First, it combines semantic and behavioural evidence within a modular decision framework. Second, it treats epistemic uncertainty as an operational control variable rather than only as a diagnostic output. Third, it links predictive risk and uncertainty to an explicit triage policy that governs analyst escalation and automatic containment.

1.1. Contributions

This subsection summarises the concrete technical contributions of the paper.

This work makes the following key contributions:

- We propose SABRE, an uncertainty-aware modular ransomware detection framework that combines a semantic detector, a behavioural detector, and a score-level decision layer with triage support.
- We operationalise epistemic uncertainty through Monte Carlo dropout, enabling each detector to output both a predictive score and an uncertainty estimate.
- We introduce an interpretable triage policy governed by a risk threshold τ and an uncertainty threshold κ, enabling calibrated decisions among allow, analyst escalation, and automatic containment.
- We examine the effect of score-space decision modelling, including CTGAN-based score augmentation and uncertainty-aware inputs, through ablation studies.
- We provide interpretability analyses based on permutation importance, gradient-based sensitivity, and counterfactual perturbations to characterise the decision behaviour of the semantic and behavioural components.

Paper structure

The remainder of this paper is organised as follows. Section 2 provides background on ransomware detection and motivates the need for adaptive, uncertainty-aware defence under concept drift and adversarial evolution. Section 3 reviews related work on ransomware detection, uncertainty quantification, and explainable security analytics, highlighting limitations that motivate the proposed framework. Section 4 presents the SABRE framework, including the semantic and behavioural components, score-level decision modelling, uncertainty estimation via Monte Carlo Dropout, and the uncertainty-aware triage policy. Section 7 formalises the threat model and adversarial assumptions. Section 8 describes the experimental setup, including datasets, feature construction, model training, and evaluation protocols. Section 9 reports empirical results and analyses the behaviour of the uncertainty-aware triage policy under different operating regimes. Section 10 presents explainability and interpretability analyses based on feature sensitivity

and counterfactual reasoning. Section 11 reports ablation studies examining the contribution of key system components. Section 12 discusses the main empirical findings and operational implications. Section 13 outlines limitations and directions for future work. Finally, Section 14 concludes the paper.

2. Background

This section provides background on ransomware detection, uncertainty-aware machine learning, and the operational motivation for uncertainty-guided triage.

Ransomware has become a dominant form of cybercrime, driven by increasingly sophisticated techniques for evasion, polymorphism, and operational stealth. Traditional approaches to ransomware detection have relied on signature matching, heuristic rule sets, or simple statistical anomaly detection. Although effective for well-known threats, such approaches degrade rapidly in the presence of obfuscation, encryption, behavioural drift, and previously unseen attack families. As enterprise systems evolve and ransomware variants diversify, detection frameworks must adapt to changing distributions rather than relying on static assumptions Fernando et al. (2020); Razaulla et al. (2023).

Machine learning has emerged as a powerful tool for ransomware detection, leveraging system call traces, API telemetry, filesystem access patterns, and related forensic signals Zahoora et al. (2022); Alraizza and Algarni (2023); Alraizza, Algarni and Alrayzah (2025). Classical models and deep neural networks have shown promise in modelling such behavioural and semantic features. However, purely model-based detection suffers from three persistent challenges. First, data imbalance remains severe, as benign activity dominates operational telemetry. Oversampling techniques such as SMOTE and generative models such as CTGAN have been proposed to mitigate minority-class scarcity, improving training stability and representation of rare malicious events Chawla, Bowyer, Hall and Kegelmeyer (2002); Xu, Skoularidou, Cuesta-Infante and Veeramachaneni (2019). Second, semantic modelling using transformers and other embedding-based architectures has improved contextual understanding of logs and structured artefacts, yet these models often lack robustness under concept drift and out-of-distribution (OOD) conditions Fernando et al. (2020). Third, most existing machine learning detectors provide only point predictions and do not quantify predictive uncertainty, making it difficult to determine whether a model's output is reliable enough to support automated response decisions Gal and Ghahramani (2016); Gawlikowski, Tassi, Ali, Lee, Humt, Feng, Kruspe, Triebel, Jung, Roscher, Shahzad, Yang, Bamler and Zhu (2023).

Recent work in cybersecurity has begun to explore uncertainty quantification as a means of addressing model unreliability in deployment Gal and Ghahramani (2016); Gawlikowski et al. (2023); Kabuye, Issac, Yumlembam and Neera (2025). Techniques such as Monte Carlo Dropout and deep ensemble variance estimation allow detectors to express confidence levels alongside predicted probabilities. This is especially important in high-stakes applications where false positives incur operational cost and false negatives may cause severe damage. Nevertheless, most existing cybersecurity systems use uncertainty only as an auxiliary diagnostic tool rather than integrating it into a structured triage or escalation policy. As a result, model uncertainty is rarely incorporated into actionable decision-making and is seldom used to determine when suspicious activity should be escalated to a human analyst.

Parallel to these developments, modular detection architectures have become increasingly attractive because they allow different components to specialise in complementary evidence sources, such as static representations and runtime telemetry. Such modularity can improve interpretability, simplify system design, and support flexible deployment under heterogeneous data conditions. However, many existing systems still treat model combination as a conventional ensemble problem and do not connect uncertainty estimates to explicit operational actions.

In summary, prior research has advanced data augmentation, semantic embeddings, behavioural modelling, and explainability techniques, however, these efforts often address each challenge in isolation. Existing systems typically lack (i) modular detectors specialising in different evidence channels, (ii) principled uncertainty quantification at the component level, (iii) a decision mechanism that uses both predictive risk and uncertainty, and (iv) a structured triage policy linking model confidence to autonomous or human-in-the-loop decisions. Prior surveys of ransomware detection research highlight these fragmented efforts and underscore gaps in unified, operationally grounded frameworks for addressing challenges such as concept drift, uncertainty quantification, and integrated decision policies Ispahany et al. (2024). These gaps motivate SABRE as an uncertainty-aware modular framework for adaptive ransomware detection under drift, heterogeneous evidence, and operational constraints.

While prior surveys comprehensively catalogue ransomware detection techniques, they do not propose operational mechanisms that bind predictive uncertainty to executable response policies. SABRE addresses this gap by embedding uncertainty directly into a triage process that governs analyst escalation and automatic containment.

3. Related Work

This section reviews prior work on ransomware detection, data imbalance handling, semantic modelling, and uncertainty-aware security analytics, and then positions SABRE relative to these strands of research.

Research on ransomware detection intersects multiple areas including behavioural modelling, semantic analysis, uncertainty-aware machine learning, and modular detection architectures. For this paper, the most relevant distinction is whether uncertainty is merely reported as a confidence indicator or is explicitly used to govern downstream action. Prior work on uncertainty-aware ransomware and intrusion detection typically treats uncertainty as a diagnostic signal: uncertainty estimates are reported to aid interpretation, calibration analysis, or analyst awareness, however, they do not explicitly constrain system behaviour. In such settings, uncertainty remains an auxiliary output rather than a governing variable in the decision process. In contrast, SABRE treats epistemic uncertainty as an operational decision variable. Uncertainty is incorporated into the triage policy and directly affects whether the system may automatically contain a process, escalate to human analysts, or allow execution.

3.1. Machine Learning for Ransomware Detection

Behavioural ransomware detectors traditionally rely on handcrafted indicators such as entropy, API counts, I/O throughput, or memory access volatility. These features have been used in conjunction with classical machine learning models such as SVMs and random forests, but such approaches struggle to generalise to zero-day variants and polymorphic families. Recent ransomware detection studies have combined feature selection, optimization, and explainability to improve detection performance in memory-forensics settings Wu, Liu, Zhang, Yi and Yi (2026); however, these approaches generally do not operationalise predictive uncertainty through an explicit triage policy for analyst escalation and autonomous containment. Recent deep learning systems extend this line of work by leveraging system calls, registry traces, and log sequences. Transformer-based architectures Vaswani, Shazeer, Parmar, Uszkoreit, Jones, Gomez, Kaiser and Polosukhin (2017) and LLM-derived embeddings have also been explored for intrusion detection and semantic log analysis Tian, Zhang, Liu, Wang, Wu, Zhu, Zhang, Zhang, Yuan, Mao and Kim (2025); Zhang, Wu, London and Tenney (2025a); Patel, Yeh and Gondhalekar (2024). However, these systems generally operate as monolithic detectors and produce deterministic outputs without quantifying epistemic uncertainty or providing structured mechanisms for human escalation. SABRE differs from these approaches by combining semantic and behavioural evidence with explicit uncertainty estimates and a downstream triage policy.

3.2. Data Augmentation and Imbalance Handling

Imbalanced datasets remain a pervasive challenge in ransomware detection because benign activity vastly outweighs malicious traces. Oversampling approaches such as SMOTE Chawla et al. (2002) and generative tabular models such as CTGAN Xu et al. (2019); Ashrapov (2020) have been widely adopted to mitigate class imbalance. Such approaches have proven effective in cybersecurity domains, including IoT botnet detection Habibi, Chemmakha and Lazaar (2023) and enterprise anomaly detection Chaithanya and Brahmananda (2022). While these methods improve training stability, they are typically employed only at the data preprocessing stage and are rarely integrated into multi-stage detection pipelines. In SABRE, CTGAN is applied in score space rather than raw feature space, with the aim of improving coverage of rare decision-region configurations seen by the decision layer.

3.3. Semantic Modelling and Log Understanding

LLMs and transformer-based encoders have demonstrated strong performance in security log analysis and anomaly detection Tian et al. (2025); Zhang et al. (2025a). Systems such as LLM-LADE Zhang, Li, Zhang, Ye, Hu and Yan (2025b) combine BERT-style embeddings with SHAP explanations Lundberg and Lee (2017) to provide interpretable semantic anomaly detection. However, these systems focus primarily on semantic or log-based analysis and do not jointly study runtime behavioural telemetry and uncertainty-guided response decisions in ransomware-specific settings. SABRE differs by placing semantic representations alongside behavioural evidence within a modular detection-and-triage framework.

3.4. Uncertainty Quantification in Cybersecurity

There is growing interest in incorporating uncertainty estimation into deep learning models for security Kabuye et al. (2025). Methods such as Monte Carlo Dropout, deep ensembles, and Bayesian neural networks have been used to improve robustness and identify unreliable predictions. However, existing systems typically use uncertainty as an auxiliary diagnostic rather than as a structured driver of decision policies. Few works integrate uncertainty thresholds

Approach class	Semantic + behavioural evidence	Uncertainty used operationally	Explicit triage policy	Interpretability analysis
Classical ransomware ML	Partial	No	No	Limited
Deep behavioural detectors	Limited	No	No	Limited
Uncertainty-aware security models	Partial	Usually diagnostic only	No	Partial
SABRE (this work)	Complementary	Yes	Yes	Yes

Table 1
Positioning of SABRE relative to representative classes of prior ransomware detection approaches.

into automated response pipelines or calibrate model behaviour using formal triage criteria. SABRE advances this direction by embedding uncertainty estimates directly into an operational triage policy governed by risk and uncertainty thresholds (τ, κ), thereby enabling selective automation under uncertainty.

3.5. Modular Detection Architectures

Modular detection architectures have been explored in intrusion detection and adaptive cybersecurity because they allow different models or subsystems to specialise in complementary evidence sources and decision roles. Such designs can improve interpretability, permit staged reasoning, and support more flexible deployment than monolithic classifiers. However, prior modular systems typically lack (i) component-level uncertainty estimation, (ii) uncertainty-aware decision fusion, and (iii) operational triage policies linking uncertainty and risk to automated or human-in-the-loop decisions. SABRE addresses this gap by combining semantic and behavioural detection components with uncertainty-aware decision support for containment, escalation, or allowance. This positions SABRE as a practical modular ransomware detection framework rather than as a conventional ensemble alone.

Relative to prior ransomware detection studies, the novelty of SABRE is threefold. First, it combines semantic and behavioural evidence within a modular decision framework. Second, it treats epistemic uncertainty as an operational control variable rather than as a diagnostic output alone. Third, it links predictive risk and uncertainty to an explicit triage policy that governs automatic containment versus analyst escalation.

4. SABRE Framework

4.1. System Overview

This section describes the SABRE framework, including its modular architecture, uncertainty estimation procedure, score-level decision layer, and uncertainty-aware triage mechanism.

SABRE is an uncertainty-aware modular framework designed to support ransomware detection under concept drift and behavioural polymorphism. The framework consists of five stages: (i) feature extraction, (ii) augmentation, (iii) detector training with Monte Carlo Dropout, (iv) score-level decision modelling, and (v) uncertainty-aware triage. The operational pipeline is illustrated in Fig. 1.

The framework contains two specialised detection components. The semantic component operates on PE-derived representations, while the behavioural component operates on runtime telemetry features. Each component produces both a predictive mean score and an epistemic uncertainty estimate. These outputs are passed to a lightweight decision layer that computes a fused ransomware risk score. A triage policy then determines whether the sample should be automatically contained, escalated to a human analyst, or allowed.

Unlike conventional uncertainty-aware detectors, which use predictive uncertainty primarily for post-hoc interpretation, SABRE embeds uncertainty directly into the decision logic. The thresholds κ and κ_{low} act as explicit uncertainty budgets, defining regions in which automated action is permitted or prohibited. As a result, samples with high predicted risk but elevated epistemic uncertainty are prevented from triggering automatic containment and are instead routed to human analysts. The modular structure of SABRE also allows the triage policy to be recalibrated without retraining the base detectors.

4.2. Score-Level Decision Layer

The decision layer operates on a four-dimensional score vector that concatenates the predictive means and uncertainties produced by the semantic and behavioural detectors:

$$\mathbf{s} = \begin{bmatrix} \bar{p}_{\text{sem}} \\ \bar{p}_{\text{beh}} \\ \sigma_{\text{sem}} \\ \sigma_{\text{beh}} \end{bmatrix}. \tag{1}$$

Because the semantic and behavioural datasets are heterogeneous and are not paired at the per-sample level, joint score tuples are constructed by label-aligned pairing. In particular, semantic score tuples are paired with behavioural score tuples of the same class label, and vice versa, to form a score-space training set for the decision layer. This allows the decision model to learn from complementary evidence sources without requiring matched static and runtime observations of the same executable instance.

When score-space augmentation is used, CTGAN is applied only to the training split of this score representation to improve coverage of rare decision-region configurations. A lightweight multilayer perceptron then maps the score vector to a fused ransomware risk estimate:

$$\hat{p} = f(\mathbf{s}). \tag{2}$$

In the reported implementation, the decision layer additionally supports post-hoc temperature scaling on the validation split to improve calibration of the fused risk score. This score-level formulation is deliberately lightweight, allowing the decision layer to use both predictive strength and detector-level uncertainty information while remaining interpretable and easy to recalibrate.

4.3. Uncertainty Aggregation

To ensure conservative behaviour, component-level uncertainties are aggregated using a max-operator:

$$\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}}). \tag{3}$$

This design ensures that elevated uncertainty in either evidence channel propagates to the final triage decision. Algorithms 1–4 summarise the training, inference, score-level decision modelling, and deployment-time triage logic of SABRE.

Algorithm 1 SABRE: Training and Deployment Pipeline

Input (training): Raw semantic data D_{sem}, behavioural logs D_{beh}
Input (deployment): New sample $x_{\text{sem}}, x_{\text{beh}}$
Output (deployment): Decision $\in$ {AUTO-CONTAIN, ESCALATE, ALLOW}

// Phase 1: Feature extraction
Extract semantic embeddings $\mathbf{z}_i$ from D_{sem} using encoder $\mathcal{E}(\cdot)$
Extract behavioural feature vectors $\mathbf{b}_i$ from D_{beh} via sliding-window statistics

// Phase 2: Feature-level augmentation
Apply SMOTE to $\{\mathbf{z}_i\}$ and $\{\mathbf{b}_i\}$ independently to address class imbalance

// Phase 3: Train base detectors with MC Dropout
Train semantic detector A_{sem} on $\{\mathbf{z}_i\}$
Train behavioural detector A_{beh} on $\{\mathbf{b}_i\}$

// Phase 4: Score-level augmentation and decision layer
for each semantic sample i **do**
    Compute $(\bar{p}_{\text{sem}}^{(i)}, \sigma_{\text{sem}}^{(i)})$ via MC Dropout
    Pair with a behavioural score tuple of the same class label
end for
for each behavioural sample j **do**
    Compute $(\bar{p}_{\text{beh}}^{(j)}, \sigma_{\text{beh}}^{(j)})$ via MC Dropout
    Pair with a semantic score tuple of the same class label
end for
Form score tuples $\mathbf{s} = [\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}]$
Apply CTGAN to the training split of the score tuples and obtain augmented scores $\tilde{D}_S$ (Alg. 3)
Train decision layer A_{dec} on $D_S \cup \tilde{D}_S$

// Phase 5: Deployment-time inference and triage
Obtain $(\bar{p}_{\text{sem}}, \sigma_{\text{sem}})$ for x_{sem} via MC Dropout
Obtain $(\bar{p}_{\text{beh}}, \sigma_{\text{beh}})$ for x_{beh} via MC Dropout
Form $\mathbf{s} = [\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}]$
Compute fused risk $\hat{p} = A_{\text{dec}}(\mathbf{s})$
Compute aggregated uncertainty $\sigma_{\max} = \max(\sigma_{\text{sem}}, \sigma_{\text{beh}})$
Apply triage policy using thresholds $(\tau, \kappa, \tau_{\text{high}}, \kappa_{\text{low}})$ (Alg. 4)

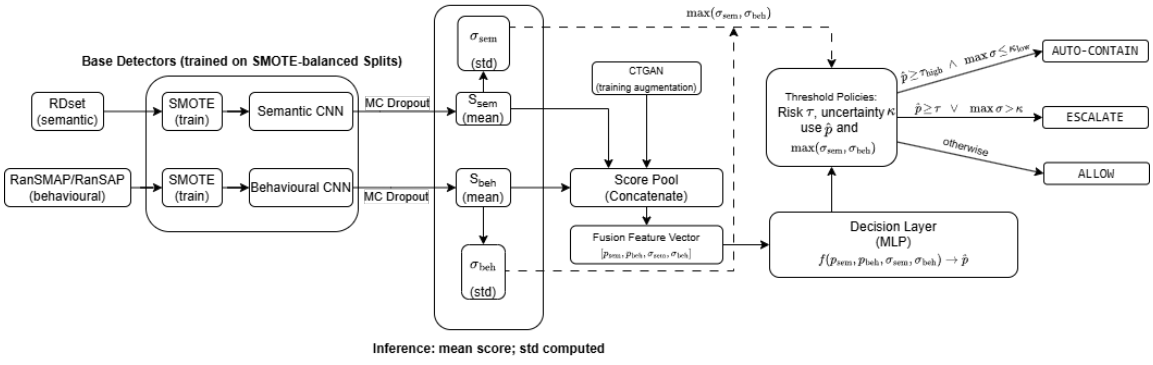


Figure 1: Operational pipeline of SABRE. The semantic and behavioural detectors, each equipped with Monte Carlo Dropout, produce predictive mean scores and uncertainty estimates. These outputs are combined in a score-level decision layer to produce a fused risk score $\hat{p}$. A triage policy then uses a risk threshold τ and an uncertainty threshold κ, together with stricter containment thresholds τ_{high} and κ_{low}, to assign one of three actions: *AUTO-CONTAIN*, *ESCALATE*, or *ALLOW*.

Algorithm 2 MC Dropout Uncertainty Estimation for a Neural Detector

Input: Trained neural detector A with dropout layers, sample x, number of passes T
Output: Predictive mean $\bar{p}$, epistemic uncertainty σ
Set A to evaluation mode
Activate dropout layers for inference
for $t = 1$ to T **do**
    $p_t \leftarrow A(x)$
end for
Compute predictive mean:

$$\bar{p} = \frac{1}{T} \sum_{t=1}^{T} p_t$$

Compute predictive standard deviation:

$$\sigma = \sqrt{\frac{1}{T} \sum_{t=1}^{T} (p_t - \bar{p})^2}$$

return $(\bar{p}, \sigma)$

Algorithm 3 Score-Level CTGAN Augmentation for Decision-Layer Training

Input: Score tuples $\mathcal{D}_S = \{(\mathbf{s}_i, y_i)\}$
Output: Augmented score dataset $\tilde{\mathcal{D}}_S$
// Step 1: Build score-space training tuples
Construct label-aligned score tuples $\mathbf{s}_i = [\bar{p}^{(i)}_{\text{sem}}, \bar{p}^{(i)}_{\text{beh}}, \sigma^{(i)}_{\text{sem}}, \sigma^{(i)}_{\text{beh}}]$
Split $\mathcal{D}_S$ into training and validation subsets
// Step 2: Train CTGAN in score space
Train CTGAN on the training split only, conditioned on labels y_i
Generate synthetic score samples $\tilde{\mathbf{s}}_j$
Form augmented training set $\tilde{\mathcal{D}}_S = \{(\tilde{\mathbf{s}}_j, y_j)\}$
// Step 3: Train decision layer
Train decision layer A_{dec} on the original and synthetic training tuples
Optionally apply post-hoc temperature scaling on the validation split
return $\tilde{\mathcal{D}}_S$

Algorithm 4 Uncertainty-Aware Triage Policy

Input: Fused risk $\hat{p}$, component uncertainties $\sigma_{\text{sem}}, \sigma_{\text{beh}}$
Input: Risk thresholds τ, τ_{high}; uncertainty thresholds κ, κ_{low}
Output: Decision $\in$ {AUTO-CONTAIN, ESCALATE, ALLOW}
Compute aggregated uncertainty $\sigma_{\max} \leftarrow \max(\sigma_{\text{sem}}, \sigma_{\text{beh}})$
if $\hat{p} \geq \tau_{\text{high}}$ **and** $\sigma_{\max} \leq \kappa_{\text{low}}$ **then**
    return AUTO-CONTAIN
else if $\hat{p} \geq \tau$ **or** $\sigma_{\max} > \kappa$ **then**
    return ESCALATE
else
    return ALLOW
end if

5. Uncertainty-Aware Triage Policy

This section formalises the uncertainty-aware triage policy used by SABRE and explains how fused risk and predictive uncertainty are translated into operational actions.

SABRE can be interpreted as a decision-support framework in which epistemic uncertainty constrains autonomous action. Rather than optimising predictive accuracy alone, the framework is designed to support operational utility by regulating when automatic containment is permitted and when human oversight is required. In this formulation, epistemic uncertainty is treated not merely as a diagnostic by-product of prediction, but as an explicit decision variable that governs whether the system may act autonomously or must defer to analysts.

Given a fused ransomware risk score $\hat{p} \in [0, 1]$ and an aggregated epistemic uncertainty $\sigma_{\max} \in [0, 1]$, the triage layer applies a two-threshold policy regulated by: (i) a risk threshold τ and (ii) an uncertainty budget κ. For safer autonomous containment, stricter thresholds τ_{high} and κ_{low} define a high-confidence operating region.

5.1. Policy Design and Thresholds

This subsection defines the threshold structure used by the triage policy.

Axis-aligned thresholds are deliberately chosen over learned non-linear decision boundaries to preserve interpretability, auditability, and policy stability under distribution shift, properties that are important in operational security settings.

The policy uses two pairs of thresholds: (i) (τ, κ) defining the general escalation boundary, and (ii) $(\tau_{\text{high}}, \kappa_{\text{low}})$ defining a stricter region in which autonomous containment is permitted.

The thresholds satisfy

$$0 \le \tau \le \tau_{\text{high}} \le 1, \qquad 0 \le \kappa_{\text{low}} \le \kappa \le 1, \tag{4}$$

so that automatic containment is always a stricter decision than escalation.

Intuitively, τ controls the minimum risk level that warrants intervention, while κ constrains how much epistemic uncertainty is tolerated before human escalation is required. The stricter pair $(\tau_{\text{high}}, \kappa_{\text{low}})$ enforces a conservative automation contract: containment is allowed only when the fused risk is very high and uncertainty is sufficiently low.

5.2. Decision Rules

Let $\sigma_{\max}$ denote the aggregated epistemic uncertainty computed from the semantic and behavioural components. The triage layer maps each sample to one of three actions: ALLOW, ESCALATE, or AUTO-CONTAIN, according to:

$$\text{AUTO-CONTAIN} \quad \text{if} \quad \hat{p} \ge \tau_{\text{high}} \ \wedge \ \sigma_{\max} \le \kappa_{\text{low}}, \tag{5}$$

$$\text{ESCALATE} \quad \text{if} \quad \hat{p} \ge \tau \ \vee \ \sigma_{\max} > \kappa, \tag{6}$$

$$\text{ALLOW} \quad \text{otherwise.} \tag{7}$$

5.3. Operational Interpretation

The policy defines a calibrated balance between autonomous defence and analyst workload. High-risk, low-uncertainty samples fall into the **AUTO-CONTAIN** region (Eq. 5), enabling immediate response when the system is confident. Samples that are either sufficiently risky or epistemically uncertain are routed to **ESCALATE** (Eq. 6), preventing overconfident automation under distribution shift, concept drift, or ambiguous evidence. Remaining samples are **ALLOW** (Eq. 7), corresponding to low-risk decisions made with acceptable confidence.

From a deployment perspective, τ and κ act as user-tunable controls that trade off false escalations versus false allowances, while $(\tau_{\text{high}}, \kappa_{\text{low}})$ provides an additional safety margin that limits automated containment to the highest-confidence cases.

6. Policy Analysis

This section provides a geometric interpretation of the triage policy in the joint risk–uncertainty space.

The triage mechanism in SABRE is governed by the joint behaviour of the fused probability $\hat{p}$ and the maximal epistemic uncertainty $\sigma_{\max}$. The policy partitions the space

$$\mathcal{X} = \left\{ (\hat{p},\, \sigma_{\max}) \in [0, 1] \times [0, 1] \right\} \tag{8}$$

into three disjoint regions corresponding to the actions ALLOW, ESCALATE, and AUTO_CONTAIN.

The decision boundaries are defined by the inequalities

$$\hat{p} \geq \tau_{\text{high}}, \qquad \sigma_{\max} \leq \kappa_{\text{low}}, \tag{9}$$

for autonomous containment, and

$$\hat{p} \geq \tau \quad \text{or} \quad \sigma_{\max} > \kappa, \tag{10}$$

for escalation. The remaining region yields the allow decision.

The geometry of the decision space is determined by the relative magnitudes of τ, τ_{high}, κ, and κ_{low}. The boundaries $\hat{p} = \tau$ and $\hat{p} = \tau_{\text{high}}$ form vertical lines, while $\sigma_{\max} = \kappa$ and $\sigma_{\max} = \kappa_{\text{low}}$ form horizontal lines. Their intersections partition the decision space into operational regions corresponding to the three triage outcomes.

The optimisation of the triage thresholds identifies threshold pairs that shift these boundaries. Increasing τ reduces the size of the escalation region; however, without a corresponding adjustment to κ, it may also enlarge the allow region. This trade-off reflects the central role of joint risk–uncertainty thresholding in balancing automated action, escalation volume, and operational conservatism.

7. Threat Model

This section describes the operational threat model and clarifies the assumptions under which SABRE is intended to support ransomware triage.

We consider an adversary capable of deploying ransomware samples whose static structure, runtime behaviour, or both may vary across families and executions. The defender observes two forms of evidence: a semantic representation $\mathbf{z}_i \in \mathbb{R}^{d_s}$ derived from static artefacts and a behavioural representation $\mathbf{b}_i \in \mathbb{R}^{d_b}$ derived from temporal I/O activity. The threat model assumes that an attacker may attempt to alter one or both evidence channels, but cannot arbitrarily manipulate them without affecting executable structure, runtime feasibility, or attack effectiveness.

The main adversarial objective considered in this work is evasion under functional constraints. In particular, an attacker may seek to reduce the model's estimated risk or increase uncertainty while preserving malicious behaviour. On the semantic side, such manipulation may involve modifying static descriptors or PE-level artefacts. On the behavioural side, the attacker may attempt mimicry, that is, altering observable execution patterns so that malicious activity appears closer to benign runtime behaviour.

For the behavioural setting, we denote the class-conditional feature distributions by

$$p(\mathbf{b} \mid y = 1), \qquad p(\mathbf{b} \mid y = 0), \tag{11}$$

where behavioural mimicry attempts to reduce separability between malicious and benign telemetry. In practice, such manipulations may also increase ambiguity in the observed evidence, which is relevant because the triage mechanism uses epistemic uncertainty as part of the final operational decision.

We do not assume that the attacker is fully unaware of the system design; however, the analysis does assume that successful evasion is constrained by the need to preserve executable validity and malicious intent. Under this assumption, SABRE is designed to behave conservatively: samples that appear high-risk but also uncertain are redirected to human analysts rather than automatically contained or allowed. Accordingly, the threat model does not claim formal robustness guarantees against all adversarial strategies. Instead, it motivates the practical role of uncertainty-aware triage as a safeguard against overconfident automated action under ambiguous or potentially adversarial conditions.

8. Experimental Setup

This section describes the datasets, feature construction, model training procedure, and evaluation protocol used to study the semantic, behavioural, and uncertainty-aware triage components of SABRE.

8.1. Datasets

Experiments are conducted on publicly available ransomware datasets with complementary observational scopes. RDset provides static PE-derived metadata and is used to evaluate the semantic component. RanSMAP and RanSAP

provide behavioural telemetry extracted from system-level I/O activity and are used to evaluate the behavioural component. These datasets are heterogeneous and are not paired at the per-sample level. Accordingly, this study does not claim joint multimodal learning over matched semantic and behavioural observations of the same executable instance. Instead, SABRE is evaluated as a modular framework in which semantic and behavioural components are analysed on complementary datasets and connected through a common uncertainty-aware decision formulation.

RanSAP Hirano, Hodota and Kobayashi (2022) is a hypervisor-level behavioural dataset consisting of low-level storage access patterns only, comprising approximately 200 million I/O events aggregated into fixed-length temporal windows collected from multiple ransomware and benign executions. RanSMAP Hirano and Kobayashi (2025), a successor dataset, extends RanSAP by incorporating both storage and memory access patterns, yielding 11,820 execution traces across diverse hardware configurations, ransomware variants, and simultaneous benign–malicious workloads. In contrast, the dataset introduced by Mathur Mathur (2020), commonly referred to as *RDset* in subsequent literature Kabuye et al. (2025), comprises 138,047 PE samples described by 57 structured static attributes. In this work, these attributes are transformed into semantic representations via a lightweight text-based encoding and embedding procedure.

All datasets are split into disjoint training and test sets at the sample level, ensuring that no identical execution traces or binaries appear in both splits. Because the datasets capture different evidence modalities and were collected under different settings, the experiments should be interpreted as complementary evaluations of the semantic and behavioural parts of the framework rather than as a single paired multimodal benchmark.

8.2. Feature Construction and Preprocessing

SABRE operates on two complementary evidence sources: semantic representations derived from RDset and behavioural telemetry derived from RanSMAP/RanSAP.

Semantic Features (RDset). Structured metadata fields from PE headers, entropy descriptors, and resource fields are concatenated into a text-like string and encoded into fixed-length vectors using a lightweight transformer-based sentence encoder. Formally, for each instance with metadata vector $\mathbf{x}_i = \{x_{i,1}, \ldots, x_{i,d}\}$, we form

$$\tilde{t}_i = \mathcal{T}(\mathbf{x}_i) = \texttt{Join}(\texttt{str}(x_{i,1}), \ldots, \texttt{str}(x_{i,d})), \tag{12}$$

and obtain a contextual embedding

$$\mathbf{z}_i = \mathcal{E}(\tilde{t}_i) \in \mathbb{R}^k. \tag{13}$$

These semantic embeddings form the input to the semantic detector.

Behavioural Features (RanSMAP/RanSAP). System logs capturing memory activity, I/O throughput, and entropy statistics are segmented using a sliding window. For each window $\mathcal{W}_i$, we compute variance, entropy, and access-count statistics:

$$\mathrm{Var}(x) = \frac{1}{n}\sum_{j=1}^{n}(x_j - \bar{x})^2, \tag{14}$$

$$\mathcal{H}(x) = -\sum_{j=1}^{n} p(x_j)\log_2(p(x_j) + \epsilon), \tag{15}$$

$$c_k = \sum_{j=1}^{n} \mathbb{1}(x_j = k). \tag{16}$$

This yields a behavioural feature vector $\mathbf{b}_i$ for each window.

Class imbalance is mitigated using a dual augmentation strategy: (i) SMOTE at the feature level for both semantic and behavioural datasets, and (ii) CTGAN at the score level after base-detector inference. Score-level augmentation is used to enrich rare decision-region configurations seen by the decision layer without directly altering the uncertainty estimation procedure of the base detectors.

8.3. Model Training and Inference

This subsection describes the training of the semantic and behavioural detectors, the construction of score-space tuples for the decision layer, and the computation of predictive uncertainty.

Two specialised neural detectors are trained independently:

- **Semantic detector:** a CNN trained on semantic embeddings.
- **Behavioural detector:** a CNN trained on handcrafted telemetry features.

At inference time, each detector uses Monte Carlo Dropout to estimate predictive uncertainty. For an input sample x, we compute T stochastic forward passes $p_1, \dots, p_T$, yielding

$$\bar{p} = \frac{1}{T}\sum_{t=1}^{T} p_t, \qquad \sigma = \sqrt{\frac{1}{T}\sum_{t=1}^{T}(p_t - \bar{p})^2}. \tag{17}$$

For each detector we obtain $(\bar{p}_{\text{sem}}, \sigma_{\text{sem}})$ and $(\bar{p}_{\text{beh}}, \sigma_{\text{beh}})$. In the reported implementation, Monte Carlo Dropout uses $T = 10$ stochastic forward passes for both detectors.

To train the decision layer, we construct four-dimensional score tuples of the form

$$\mathbf{s} = (\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}, \sigma_{\text{sem}}, \sigma_{\text{beh}}). \tag{18}$$

Because RDset and RanSMAP/RanSAP are not paired at the sample level, these score tuples are formed by label-aligned pairing: each semantic score tuple is paired with a behavioural score tuple of the same class label, and vice versa. This produces a score-space training set for the decision layer without requiring matched static and runtime views of the same executable instance.

CTGAN augmentation is applied only to the training split of the score-space dataset. The decision-layer multilayer perceptron is then trained on the original and augmented training tuples, and optional post-hoc temperature scaling is applied on the validation split to improve calibration of the fused risk score.

In addition to predictive quality, deployment practicality depends on inference overhead. The semantic and behavioural detectors operate independently, and the decision layer uses a small score vector, so the main additional computational cost arises from Monte Carlo Dropout. This cost scales approximately linearly with the number of stochastic passes T, allowing a deployment trade-off between uncertainty precision and inference latency.

8.4. Evaluation Metrics and Policy Optimisation

This subsection defines the evaluation metrics and explains how policy thresholds are selected.

Performance is evaluated using accuracy, AUC, and confusion-matrix-derived metrics. In addition to classification quality, we explicitly measure operational cost via the escalation rate induced by the triage policy. Thresholds (τ, κ) are selected by optimising a scalarised objective that balances balanced accuracy against escalation volume.

The evaluation of SABRE is organised around three questions: (i) the discriminative performance of the individual semantic and behavioural detectors; (ii) the behaviour of the decision layer and uncertainty-aware triage mechanism; and (iii) the impact of the triage policy defined by the thresholds τ, κ, τ_{high}, and κ_{low}.

Because RDset and RanSMAP/RanSAP are heterogeneous datasets rather than matched views of the same instances, these score-space tuples are constructed at the detector-output level rather than from paired multimodal samples. Accordingly, the empirical results should be interpreted as evidence about the behaviour of the semantic and behavioural components under complementary conditions, and about the operational role of uncertainty-aware triage, rather than as a claim of end-to-end multimodal fusion over paired samples.

9. Results

This section reports the performance of the semantic and behavioural detectors, examines the behaviour of the uncertainty-aware triage policy, and analyses calibration, feature sensitivity, and policy geometry.

Unlike conventional evaluation metrics, the escalation rate induced by the triage policy represents an operational cost rather than a classification error. Escalated samples require analyst attention, introduce response latency,

Detector	Dataset	Accuracy	AUC
Semantic CNN	RDset	1.0000	1.0000
Behavioural CNN	RanSMAP	0.9872	0.6030

Table 2
Performance of the individual semantic and behavioural detectors.

and increase cognitive workload. Reductions in escalation volume at fixed recall therefore correspond directly to deployability improvements, even when headline accuracy remains unchanged.

The semantic detector receives embedding vectors $\mathbf{z}_i \in \mathbb{R}^{384}$ and produces a probability

$$p_{\text{sem}} = f_{\text{S}}(\mathbf{z}), \tag{19}$$

which achieves accuracy and AUC equal to 1.0000 on RDset. The separation in the semantic embedding space is therefore nearly perfect. This behaviour reflects the highly structured nature of PE-level metadata in RDset and should not be interpreted as a claim of general semantic dominance. Rather, RDset serves here as a high-separability semantic evaluation setting for examining how uncertainty-aware decision support behaves when static evidence is unusually strong.

The behavioural detector receives handcrafted statistical–temporal features $\mathbf{b}_i \in \mathbb{R}^{16}$ and computes

$$p_{\text{beh}} = f_{\text{B}}(\mathbf{b}), \tag{20}$$

resulting in accuracy 0.9872 and AUC 0.6030. Although the accuracy is high, the relatively modest AUC indicates limited ranking quality across thresholds. This discrepancy is consistent with the class distribution and the high-variance nature of behavioural telemetry: many samples are classified correctly at the selected operating point, yet the score distribution still provides only weak separation between malicious and benign windows.

Table 2 summarises these base-detector results.

9.1. Decision-Layer Behaviour and Uncertainty-Aware Triage

The decision layer operates on the score vector

$$\mathbf{s} = \left(\bar{p}_{\text{sem}},\, \bar{p}_{\text{beh}},\, \sigma_{\text{sem}},\, \sigma_{\text{beh}}\right), \tag{21}$$

and produces a fused risk score

$$\hat{p} = f(\mathbf{s}), \tag{22}$$

where $f(\cdot)$ denotes the CTGAN-augmented multilayer perceptron. In the reported implementation, these four-dimensional score tuples are constructed through label-aligned pairing across the semantic and behavioural datasets, as described in Section 8.

The epistemic uncertainty is summarised through the maximal uncertainty across components,

$$\sigma_{\max} = \max\left(\sigma_{\text{sem}},\, \sigma_{\text{beh}}\right), \tag{23}$$

consistent with the triage mechanism introduced earlier.

The *safety-optimal* triage configuration is obtained by maximising balanced accuracy over the decision rule and yields thresholds

$$\tau = 0.999999, \qquad \kappa = 0.458, \tag{24}$$

$$\tau_{\text{high}} = 0.99, \qquad \kappa_{\text{low}} = 0.229. \tag{25}$$

The very high value of τ reflects a safety-dominant operating point and should be interpreted as dataset- and objective-specific rather than universal. In practice, the important design choice is not the exact numeric value itself,

Policy	Dataset	Accuracy	AUC	AUTO	ESCALATE	ALLOW	(TN, FP, FN, TP)
Safety-Optimal	RDset	0.99999	1.0000	2,114	99,438	36,495	(41,323, 0, 1, 96,723)
Low-Escalation	RDset	0.99999	0.99999	2,099	99,499	36,449	(41,322, 1, 0, 96,724)
Safety-Optimal	RanSMAP	0.9406	0.5394	15,591	696,675	39,322	(172, 1,129, 43,630, 706,784)
Low-Escalation	RanSMAP	0.9405	0.5397	15,697	696,395	39,496	(171, 1,088, 43,656, 706,673)

Table 3
Decision-layer and triage performance under safety-optimal and low-escalation policies.

however, the use of a strict risk criterion together with an uncertainty constraint to prevent overconfident automated containment.

A second configuration, the *low-escalation regime*, solves the scalarised optimisation problem

$$J(\tau, \kappa) = \text{bal_acc}(\tau, \kappa) - \lambda \, \text{EscRate}(\tau, \kappa), \tag{26}$$

with $\lambda = 0.5$. Since the decision rule depends jointly on $\hat{p}$ and $\sigma_{\max}$, the optimisation landscape is non-smooth, and the solution reflects a compromise between accuracy and operational cost.

Table 3 reports the triage metrics across RDset and RanSMAP. In RDset, the decision-layer output achieves AUC 1.0000 under both regimes, with at most one misclassification. In contrast, RanSMAP remains difficult due to the weaker separability of the behavioural evidence, resulting in modest AUC and an escalation-heavy triage pattern. Accordingly, in the RanSMAP setting the practical value of the framework lies less in strong ranking performance and more in conservative uncertainty-aware routing of ambiguous cases.

9.2. Calibration, Feature Sensitivity, and Policy Geometry

Beyond aggregate performance metrics, we examine calibration behaviour, feature sensitivity, and policy geometry in the joint risk–uncertainty space. These analyses are intended to characterise how the uncertainty-aware decision process behaves operationally, rather than to claim uniformly strong discrimination across all settings.

Behavioural feature sensitivity. Figure 2 reports permutation-based importance for the behavioural detector, measured as the drop in AUC when individual telemetry features are randomly permuted. Features related to memory access variance and entropy dispersion induce the largest degradation in performance, indicating that the behavioural detector relies primarily on irregularity patterns rather than raw activity volume. This finding aligns with the counterfactual analysis in Section 10, where minimal behavioural perturbations concentrate along the same variance-driven dimensions.

Calibration of fused risk scores. To justify the use of the fused probability $\hat{p}$ as a meaningful risk estimate, we evaluate probabilistic calibration via reliability diagrams. Figure 3 compares the empirical fraction of positives to the predicted risk across bins. The observed alignment is consistent with reasonably calibrated risk estimates in the evaluated setting, although calibration quality should be interpreted with caution given the heterogeneous datasets and the weaker discriminative behaviour of the behavioural component. In the reported implementation, the decision-layer output is post-hoc calibrated using temperature scaling on the validation split. This property is relevant for downstream policy thresholds in Eqs. (9) and (10), because it supports using $\hat{p}$ as an operational risk indicator in the evaluated setting.

ROC characteristics under decision-layer scoring. Figures 4a and 4b visualise the receiver operating characteristics on RanSMAP and RDset, respectively. On RanSMAP, the decision-layer output remains constrained by the difficulty of the behavioural telemetry setting and does not approach near-perfect separability. On RDset, where the semantic detector already saturates, the decision-layer output preserves near-perfect discrimination.

Policy geometry in risk–uncertainty space. Figure 5 depicts the induced triage policy directly in the $(\hat{p}, \sigma_{\max})$ plane. The horizontal and vertical thresholds correspond to the analytical rules defined in Eqs. (9) and (10), partitioning the space into ALLOW, ESCALATE, and AUTO-CONTAIN regions. Empirically, benign samples concentrate in the low-risk, low-uncertainty region, while malicious or ambiguous samples with elevated epistemic uncertainty are

routed to escalation rather than automatic containment. This geometric interpretation clarifies how uncertainty acts as a stabilising control variable rather than a secondary score.

Collectively, these analyses suggest that the uncertainty-aware decision process is interpretable and operationally meaningful, even though discriminative performance differs substantially between the semantic and behavioural settings. The strongest empirical result remains the use of uncertainty-aware triage to route ambiguous cases conservatively, rather than uniformly strong detection performance across all datasets.

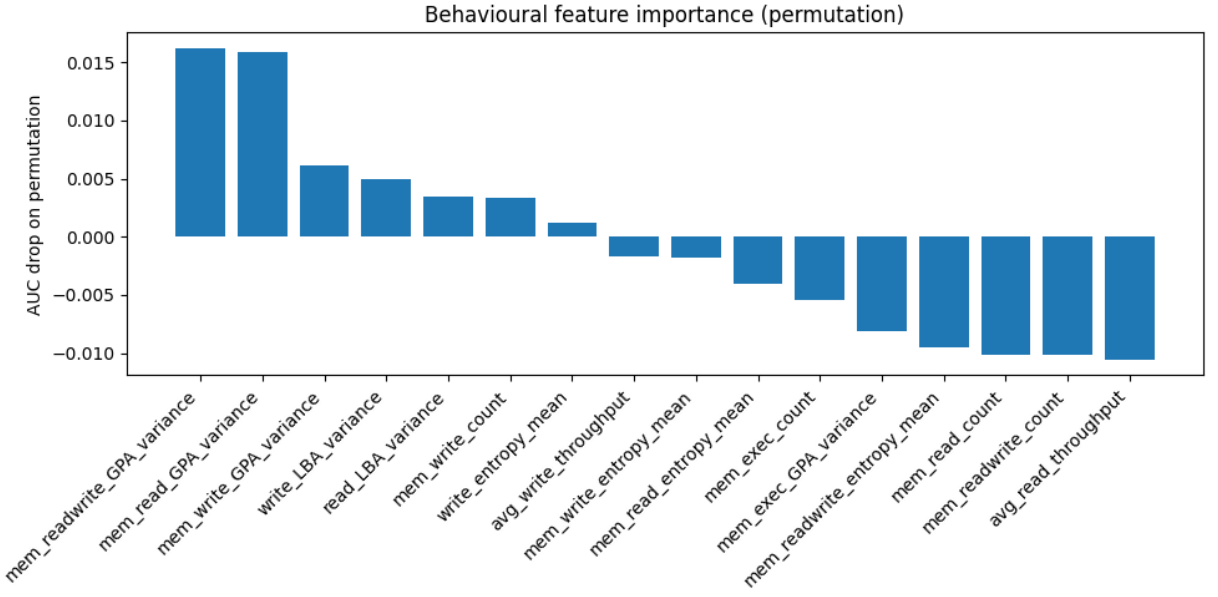


Figure 2: Permutation-based feature importance for the behavioural detector, measured as the drop in AUC after permuting each feature independently. Variance- and entropy-driven memory access statistics dominate, indicating that behavioural discrimination relies on irregularity patterns rather than raw activity volume.

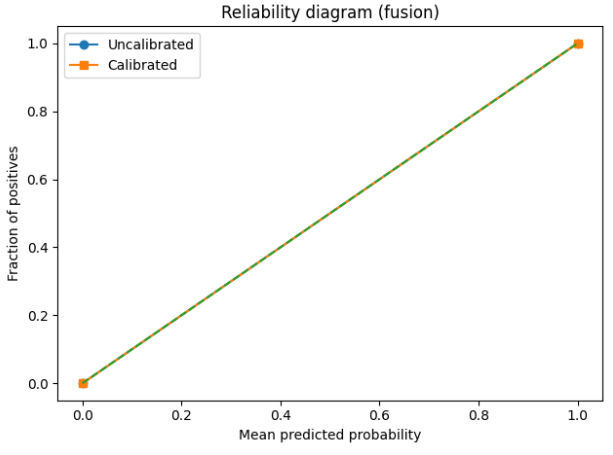
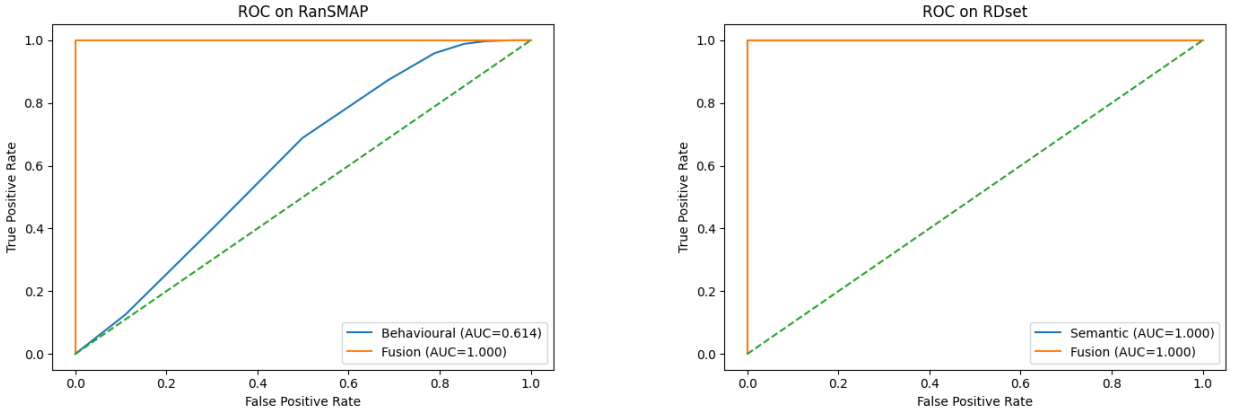


Figure 3: Reliability diagram for the fused risk score $\hat{p}$. The alignment between predicted probabilities and empirical positive rates is consistent with the use of $\hat{p}$ as an operational risk indicator for downstream triage decisions.



(a) RanSMAP (b) RDset

Figure 4: Receiver operating characteristic (ROC) curves for the evaluated settings. Left: RanSMAP, where behavioural signals are difficult to separate and decision-layer performance remains modest. Right: RDset, where the semantic detector already saturates and near-perfect discrimination is preserved.

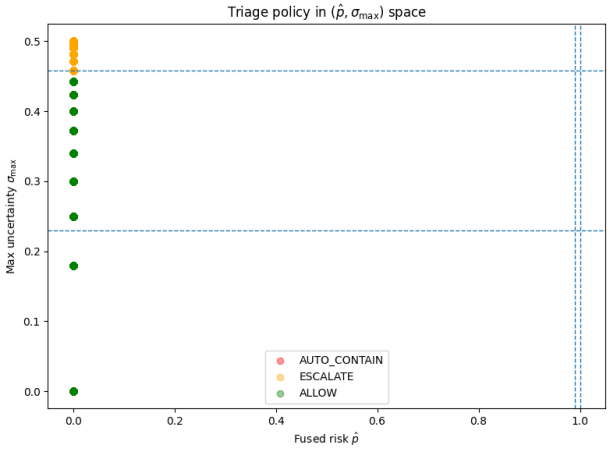


Figure 5: Triage policy geometry in the $(\hat{p}, \sigma_{\max})$ space. Horizontal and vertical thresholds correspond to the analytical decision rules in Eqs. (9) and (10), partitioning the space into ALLOW, ESCALATE, and AUTO-CONTAIN regions. Uncertainty acts as a stabilising control variable, limiting overconfident automatic containment.

10. Explainability and Interpretation

This section analyses the interpretability of the semantic and behavioural components using feature sensitivity, gradient-based saliency, and counterfactual perturbations.

We adopt counterfactual explanations as a core explainable AI mechanism because they provide local, actionable explanations by revealing the minimal semantic or behavioural changes required to alter the model's prediction. In this paper, counterfactual norms are used to characterise local decision geometry and feature sensitivity. They should not be interpreted as worst-case adversarial guarantees, a limitation discussed further in Section 13.

10.1. Behavioural Explainability via Feature Perturbation and Counterfactual Analysis

This subsection examines how the behavioural detector responds to feature perturbations and local counterfactual changes.

To understand how the behavioural detector internalises ransomware evidence, we analyse local and global sensitivities through *permutation importance* and *counterfactual explanations*. Let $\mathbf{b} \in \mathbb{R}^{d_b}$ denote the behavioural feature vector extracted from a telemetry window, and let

$$p_{\text{beh}}(\mathbf{b}) \in [0, 1] \tag{27}$$

be the behavioural detector's predictive probability.

Permutation importance assesses the global structural effect of each feature. If A_{orig} denotes the original accuracy and A_j the accuracy after independently permuting the jth feature across the dataset, its importance score is defined as

$$I_j = A_{\text{orig}} - A_j, \tag{28}$$

which measures how strongly feature b_j contributes to maintaining discriminability. The most influential features are dispersion-based quantities such as `read_LBA_variance`, `write_LBA_variance`, `mem_read_GPA_variance`, and `mem_write_GPA_variance`, suggesting that the behavioural detector relies primarily on irregularity patterns in memory and storage activity.

To complement this global sensitivity analysis, we study local model behaviour using counterfactual perturbations. Given a sample $\mathbf{b}$, we search for a minimally modified instance $\tilde{\mathbf{b}}$ within the bounded neighbourhood

$$\mathcal{N}_\epsilon(\mathbf{b}) = \left\{ \tilde{\mathbf{b}} \in \mathbb{R}^{d_b} \,:\, \|\tilde{\mathbf{b}} - \mathbf{b}\|_2 \le \epsilon \right\} \tag{29}$$

that flips the behavioural decision by satisfying

$$p_{\text{beh}}(\tilde{\mathbf{b}}) \le \theta, \tag{30}$$

where θ is a predefined low-risk target threshold used for explanation generation. In the current experiments, $\theta = 0.49$. Among all candidates satisfying (30), we select the counterfactual achieving the minimal perturbation

$$\Delta^\star = \arg \min_{\tilde{\mathbf{b}} \in \mathcal{N}_\epsilon(\mathbf{b})} \|\tilde{\mathbf{b}} - \mathbf{b}\|_2. \tag{31}$$

To describe the resulting shifts in a scale-invariant way, we define the relative perturbation for feature j as

$$\delta_j = \frac{\tilde{b}_j - b_j}{b_j} \times 100\%, \tag{32}$$

which quantifies the directional influence of counterfactual movement. Empirically, only small relative changes to key LBA and GPA variance features are sufficient to reduce high-risk behavioural scores. This behaviour is consistent with the permutation-importance analysis and indicates that dispersion-based telemetry features play a central role in local behavioural decisions.

Table 4 summarises representative behavioural counterfactual transitions.

Taken together, the behavioural permutation and counterfactual analyses suggest that the detector's decisions are driven by interpretable dispersion-related telemetry features rather than by raw activity volume alone.

Sample	$p_{\text{beh}}(\mathbf{b})$	$p_{\text{beh}}(\tilde{\mathbf{b}})$	$\|\tilde{\mathbf{b}} - \mathbf{b}\|_2^{\text{norm}}$	Most changed features (relative)
CF_1	0.70	0.30	0.12	`read_LBA_variance` (−1.6%), `mem_write_GPA_variance` (+0.6%)
CF_2	0.60	0.10	0.09	`write_LBA_variance` (−0.9%), `mem_read_GPA_variance` (−0.6%)
CF_3	0.90	0.40	0.14	`mem_write_GPA_variance` (−2.5%), `write_LBA_variance` (+1.4%)

Table 4
Representative behavioural counterfactuals generated using the minimal-perturbation formulation in Equation (31). Relative changes are computed using Equation (32).

10.2. Semantic Explainability via Embedding Sensitivity and Counterfactual Deformation

This subsection studies the interpretability of the semantic detector in embedding space.

The semantic detector operates on a dense embedding $\mathbf{z} \in \mathbb{R}^{d_s}$ derived from structured metadata and PE-header attributes. Its probabilistic output is defined as

$$p_{\text{sem}}(\mathbf{z}) \in [0, 1], \tag{33}$$

which mirrors the behavioural score in Equation (27). To characterise the internal geometry of this classifier, we analyse local sensitivity, global embedding dependence, and minimal counterfactual deformations directly in latent space.

Gradient-based semantic sensitivity. Local semantic sensitivity is quantified through gradient saliency, defined for each embedding coordinate as

$$g_j = \left| \frac{\partial p_{\text{sem}}(\mathbf{z})}{\partial z_j} \right|. \tag{34}$$

Equation (34) measures the infinitesimal influence of latent dimension j on the semantic decision boundary. Empirically, high-magnitude gradients concentrate in embedding directions associated with entropy-derived section statistics, import-table irregularities, and anomalous structural metadata, indicating that the semantic detector captures interpretable PE-level cues.

Permutation importance in embedding space. While gradient saliency captures local sensitivity, global semantic reliance is evaluated through permutation importance. Let A_{orig} denote the baseline semantic accuracy and A_j the accuracy obtained after independently permuting latent coordinate j across the evaluation set. The semantic permutation score is defined as

$$I_j^{\text{sem}} = A_{\text{orig}} - A_j, \tag{35}$$

which directly parallels the behavioural permutation measure in Equation (28). The largest values of I_j^{sem} arise from embedding components linked to structural PE irregularities and abnormal import usage. Dimensions with high permutation impact also tend to exhibit large gradient magnitudes in Equation (34), indicating consistency between local and global sensitivity views.

Semantic counterfactual deformation. To probe local interpretability, we construct semantic counterfactuals that minimally perturb the embedding while enforcing a semantic label flip. Given an original embedding $\mathbf{z}_0$, we seek a counterfactual $\mathbf{z}_{\text{cf}}$ satisfying

$$p_{\text{sem}}(\mathbf{z}_{\text{cf}}) \leq \theta_{\text{sem}}, \tag{36}$$

where θ_{sem} is a low-risk target threshold used for explanation generation. In the current experiments, $\theta_{\text{sem}} = 0.49$. The counterfactual is obtained by solving

$$\mathbf{z}_{\text{cf}} = \arg\min_{\mathbf{z}} \Big(\max\big(0, \text{logit}(\mathbf{z}) - \text{logit}(\theta_{\text{sem}})\big) \Big)^2 + \lambda \|\mathbf{z} - \mathbf{z}_0\|_2^2, \tag{37}$$

Index	$\hat{p}_{\text{sem}}(z_0)$	$\text{logit}(z_0)$	$\hat{p}_{\text{sem}}(z_{\text{cf}})$	$\text{logit}(z_{\text{cf}})$	$\|\Delta z\|_2$
102930	1.0000	32.4231	0.0000	-12.4927	2.740365
90761	1.0000	37.5973	0.0000	-17.9432	3.333000
67417	1.0000	37.4547	0.0000	-16.2387	3.335646
71097	1.0000	38.7963	0.0000	-17.5210	3.340490
123595	1.0000	36.7834	0.0000	-17.9020	3.332573

Table 5
Semantic counterfactual flips for the semantic detector. Each counterfactual embedding z_{cf} is optimised to cross a low-risk target boundary while remaining close to the original embedding z_0, consistent with the minimal-perturbation formulation in Eq. (40).

Metric	Mean	Std	$q_{0.10}$	$q_{0.90}$
$\|\Delta z\|_2$	3.2716	0.1727	3.2437	3.3395
$\|\Delta z\|_2/\|z\|_2$	0.4900	0.0283	0.4669	0.5137
Flip success to $\theta = 0.49$		100.0%		

Table 6
Summary statistics for semantic counterfactual perturbations over $N = 50$ borderline-malicious samples.

where

$$\text{logit}(\mathbf{z}) = \log\left(\frac{p_{\text{sem}}(\mathbf{z})}{1 - p_{\text{sem}}(\mathbf{z})}\right). \tag{38}$$

The optimisation is further constrained to a trust region $\|\mathbf{z}_{\text{cf}} - \mathbf{z}_0\|_2 \leq \epsilon$.

To interpret semantic displacement at the coordinate level, we define the normalised counterfactual change

$$\delta_j^{\text{sem}} = \frac{z_{\text{cf},j} - z_{0,j}}{\|\mathbf{z}_0\|_2 + \varepsilon}, \tag{39}$$

which remains well-conditioned in high-dimensional embedding space. While Eq. (37) gives the optimisation used to obtain the semantic counterfactual, Eq. (40) provides its equivalent minimal-perturbation interpretation in latent space.

$$\Delta_{\text{sem}}^{\star} = \arg\min_{\tilde{\mathbf{z}} \in \mathcal{N}_\epsilon(\mathbf{z})} \|\tilde{\mathbf{z}} - \mathbf{z}\|_2, \tag{40}$$

Geometric consistency across explainability views. Across $N = 50$ borderline-malicious samples, semantic counterfactuals consistently achieve label flips with relatively small perturbations, as visualised in Figure 6. The empirical distribution of $\|\Delta\mathbf{z}\|_2$ exhibits a mean of 3.27 with low variance (Table 6), and an observed 100% flip success rate to $\theta_{\text{sem}} = 0.49$ in this evaluated sample set. Representative sample-level transitions are reported in Table 5, which shows that high-confidence semantic predictions can be driven across the target boundary with bounded latent-space perturbations. The embedding dimensions most frequently perturbed under Equation (39) broadly coincide with those identified by high gradient saliency and large permutation impact, yielding a coherent interpretation of the semantic detector's local geometry.

Table 7 further shows that only a small subset of latent dimensions recurs across semantic counterfactuals, supporting the claim that the semantic detector relies on structured and repeatable embedding directions rather than diffuse latent variation.

10.3. Practical Interpretation and Limits

The explainability analyses above provide insight into which semantic and behavioural features influence local and global decisions. However, several limitations should be noted. First, counterfactual perturbations are computed

Latent dim	Count in Top-8	Interpretation
269	4/5	recurrent semantic direction
270	4/5	recurrent semantic direction
170	4/5	recurrent semantic direction
271	4/5	recurrent semantic direction
362	4/5	recurrent semantic direction
363	4/5	recurrent semantic direction
314	2/5	secondary direction
194	2/5	secondary direction
104	2/5	secondary direction
245	1/5	episodic direction

Table 7
Most frequently perturbed embedding dimensions across semantic counterfactuals. "Count in Top-8" reports how often the dimension appears among the highest-magnitude normalised changes δ_{sem} per sample.

in continuous feature or embedding space and therefore do not directly correspond to executable attack transformations. Second, small counterfactual norms characterise local decision sensitivity rather than worst-case adversarial vulnerability. Third, because the semantic and behavioural evaluations are conducted on heterogeneous datasets, these explainability results should be interpreted component-wise rather than as a formal joint multimodal explanation of matched samples.

Accordingly, the main role of the explainability analysis in this paper is to improve interpretability and auditability of the semantic and behavioural detectors, rather than to establish formal robustness guarantees.

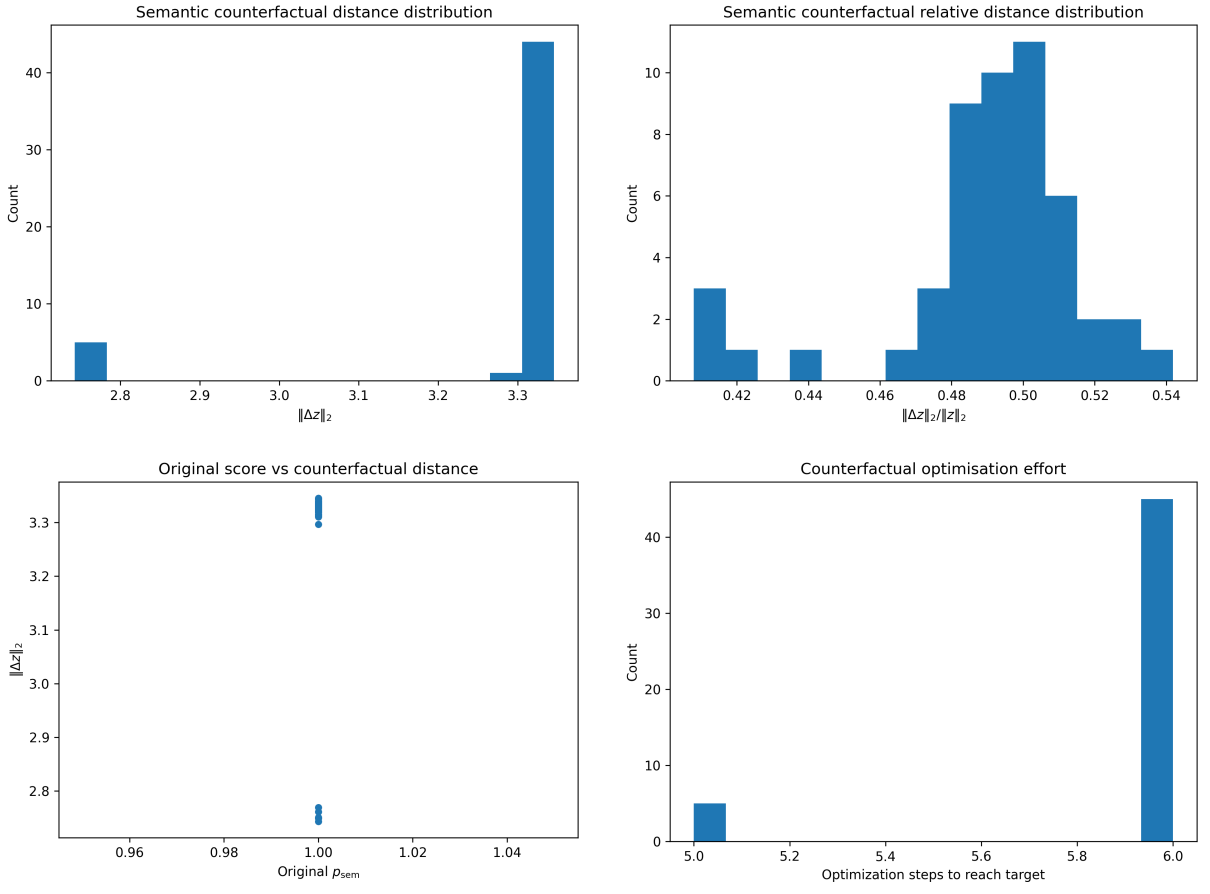


Figure 6: Semantic counterfactual diagnostics for $N = 50$ borderline-malicious samples. (Top-left) Distribution of counterfactual distance $\|\Delta z\|_2$. (Top-right) Distribution of relative distance $\|\Delta z\|_2/\|z\|_2$. (Bottom-left) Relationship between original semantic score p_{sem} and $\|\Delta z\|_2$. (Bottom-right) Optimisation steps required to reach the target $\theta = 0.49$.

11. Ablation Studies

This section evaluates the contribution of score-space augmentation, uncertainty-aware inputs, and decision-layer design.

The influence of CTGAN-based score augmentation is quantified by removing it from the decision layer. If the empirical score distribution is denoted by $\mathcal{D}_{\mathbf{s}}$, CTGAN learns a generator such that

$$\tilde{\mathcal{D}}_{\mathbf{s}} \approx p(\mathbf{s} \mid y), \tag{41}$$

producing synthetic samples that enrich the score-space training distribution. Removing CTGAN reduces the RanSMAP AUC from 0.540 to 0.514, indicating that score-space augmentation improves coverage of difficult decision regions.

Removing uncertainty inputs reduces the decision-layer input to

$$\mathbf{s}' = (\bar{p}_{\text{sem}}, \bar{p}_{\text{beh}}), \tag{42}$$

which removes Eq. (3) from the decision mechanism. As a consequence, the number of false negatives increases, since the model can no longer distinguish high-risk but uncertain samples from high-risk confident ones.

Finally, replacing the trained decision-layer MLP with the naive score mean,

$$\hat{p}_{\text{mean}} = \tfrac{1}{2}\big(\bar{p}_{\text{sem}} + \bar{p}_{\text{beh}}\big), \tag{43}$$

performs worse on RanSMAP, reflecting the limited discriminative strength of the behavioural setting and the need for a more flexible decision function.

A summary of these results is shown in Tables 8 and 9.

12. Discussion

This section interprets the empirical findings and clarifies the main practical contribution of the framework.

The experiments highlight a marked asymmetry between the discriminative behaviour of the semantic and behavioural settings. The semantic detector achieves near-saturated performance on RDset, whereas the behavioural detector on RanSMAP exhibits high accuracy at the selected operating point but only modest AUC. This contrast indicates that semantic PE-derived metadata in RDset is considerably easier to separate than behavioural telemetry in RanSMAP.

From an operational perspective, this asymmetry is important because it shows that uncertainty-aware triage is most valuable in settings where discriminative confidence is uneven across evidence sources. When evidence is strong, as in the semantic setting, the framework can support highly confident decisions. When evidence is weaker or more ambiguous, as in the behavioural setting, uncertainty-aware escalation provides a conservative safeguard against overconfident automated action.

Accordingly, the main contribution of SABRE is not a uniformly strong detection improvement across all datasets, but a transparent decision process that links predictive risk and epistemic uncertainty to analyst escalation and automatic containment. This makes the framework practically relevant even when classification performance differs substantially between semantic and behavioural settings.

The ablation studies further support this interpretation. Score-space augmentation improves the behaviour of the decision layer in difficult regions, while uncertainty inputs materially affect escalation and false-negative trade-offs. Together, these results suggest that the triage mechanism contributes most directly to deployment safety and interpretability.

13. Limitations and Future Work

Several limitations should be noted when interpreting the results of this study.

First, RDset and RanSMAP/RanSAP are heterogeneous datasets rather than matched semantic and behavioural views of the same executable instances. Accordingly, the present experiments should be interpreted as complementary evaluations of the semantic and behavioural components under different evidence settings, rather than as a strict paired

Decision-layer method	RDset AUC	RanSMAP AUC
MLP w/ CTGAN	1.0000	0.5397
MLP w/o CTGAN	0.9967	0.5142
Mean score rule (Eq. 43)	0.884	0.501

Table 8
Ablation of decision-layer strategies.

Model	RDset Esc. Rate	RanSMAP Esc. Rate	FN (RanSMAP)
Full model	0.72	0.82	43,656
No uncertainty input	0.91	0.97	58,441

Table 9
Effect of removing uncertainty inputs on triage performance.

multimodal benchmark. Future work should validate the framework on datasets that provide matched static and runtime views of the same samples.

Second, the uncertainty estimates used to compute $\sigma_{\max}$ in Eq. (3) rely on Monte Carlo Dropout sampling. While practical, this estimator may under-approximate true epistemic uncertainty in regions where behavioural data exhibit non-stationary distributions. Bayesian neural networks or deep ensembles could provide more calibrated uncertainty at higher computational cost.

Third, the triage policy described in Eqs. (9) and (10) uses axis-aligned thresholds. Although this choice improves interpretability and auditability, policy performance may depend on the selected operating point. In particular, the very strict threshold values observed in the safety-oriented regime are dataset- and objective-specific. Future work should therefore include broader sensitivity analysis and cross-dataset threshold stability evaluation.

Fourth, the explainability analyses rely partly on counterfactual perturbations in continuous feature or embedding space. These perturbations provide insight into local decision geometry, however, they do not constitute formal adversarial guarantees and do not directly model all executable constraints or runtime feasibility conditions.

Finally, the behavioural telemetry in RanSMAP remains difficult to separate, as reflected in the modest AUC values. Future work could investigate richer temporal architectures, improved calibration procedures, and evaluation under more realistic adversarial scenarios to strengthen behavioural discriminability and deployment credibility.

Despite these limitations, the modular structure of SABRE makes it possible to update the decision policy and replace individual detection components without redesigning the overall triage framework.

14. Conclusion

This paper presented SABRE, an uncertainty-aware modular framework for adaptive ransomware detection. The framework combines semantic and behavioural detectors with a lightweight decision layer that uses predictive risk and epistemic uncertainty to support triage among automatic containment, analyst escalation, and allowance.

The empirical results show that the semantic and behavioural settings differ substantially in discriminative difficulty: RDset provides a high-separability semantic setting, whereas RanSMAP remains substantially more challenging from a behavioural perspective. In this context, the main practical contribution of SABRE is not a uniformly strong performance gain across all settings, but a transparent uncertainty-aware decision process that routes ambiguous cases conservatively.

The ablation and explainability analyses further indicate that score-space augmentation, uncertainty-aware inputs, and interpretable feature sensitivity all contribute to the operational value of the framework. Overall, the results support uncertainty-aware triage as a useful deployment mechanism for ransomware detection in settings where predictive confidence is uneven and the cost of incorrect automated action is high.

Future work is to validate the framework on matched multimodal datasets, broaden threshold stability analysis, and strengthen behavioural evaluation under a more realistic adversarial and deployment settings.

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